

Mathematical studies of one-dimensional non-Hermitian topological photonic crystals

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What is a photonic crystal?

- A **photonic crystal** is a structure where the refractive index changes periodically.
- Certain arrangements create band gaps, where particular frequencies cannot enter and are reflected.
- Applications in *telecommunications, lasers, sensors, solar cells, LEDs* and all other fields of **optical engineering**!

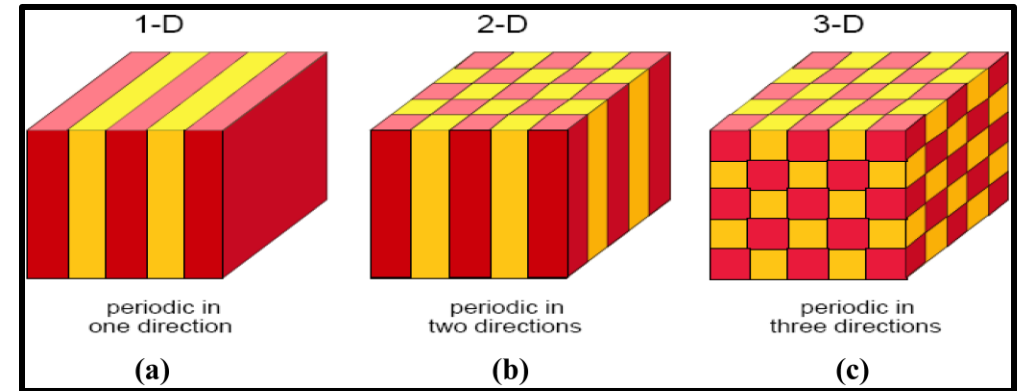
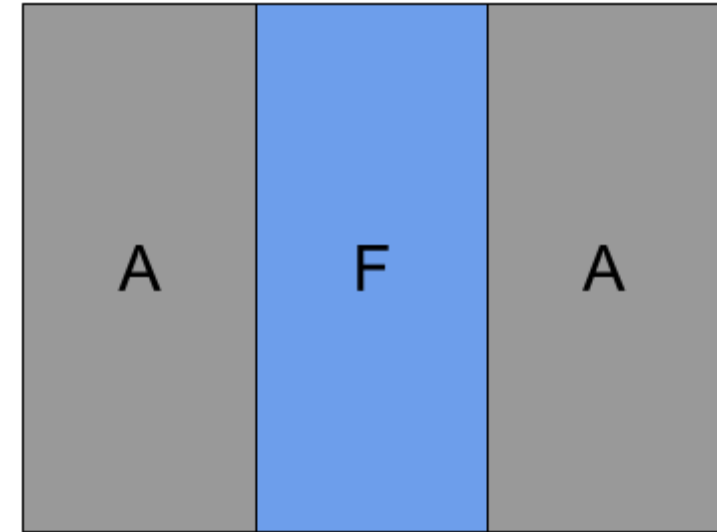


Fig. 1 – Photonic Crystal Ring Resonator Based Optical Filters
Authors: S. Robinson and R. Nakkeeran

Problem Setup

- We consider a crystal with 1D periodicity.
- The repeating *unit cell* is defined by 3 layers:
 - Layer 1: Anisotropic layer 1
 - Layer 2: Ferromagnetic layer 1
 - Layer 3: Anisotropic layer 2



*A diagram of the crystal setup.
The length of the unit cell is
defined to be 1.*

Mathematical Foundation (1/3)

- Consider the direction of propagation to be the **z** direction
- Then the permittivity and permeability tensors will vary in **z**.

$$\epsilon(z) = \begin{pmatrix} \epsilon_{xx}(z) & \epsilon_{xy}(z) & 0 \\ \epsilon_{xy}(z) & \epsilon_{yy}(z) & 0 \\ 0 & 0 & \epsilon_{zz}(z) \end{pmatrix}, \quad \mu(z) = \begin{pmatrix} \mu_{xx}(z) & \mu_{xy}(z) & 0 \\ \mu_{xy}(z) & \mu_{yy}(z) & 0 \\ 0 & 0 & \mu_{zz}(z) \end{pmatrix}$$

- The tensors will be periodic over the fundamental period of the unit cell (previously defined to be 1)

$$\epsilon(z + 1) = \epsilon(z), \quad \mu(z + 1) = \mu(z)$$

Mathematical Foundation (2/3)

- The contents of these tensors depend on the material's parameters and rotations.

$$\boldsymbol{\epsilon}_A(\delta, \varphi) = \begin{pmatrix} \epsilon_0 + \delta \cos(2\varphi) & \delta \sin(2\varphi) & 0 \\ \delta \sin(2\varphi) & \epsilon_0 - \delta \cos(2\varphi) & 0 \\ 0 & 0 & \epsilon_{zz} \end{pmatrix}, \quad \boldsymbol{\mu}_A = I.$$

$$\boldsymbol{\epsilon}_F(\alpha) = \begin{pmatrix} \tilde{\epsilon}_0 & i\alpha & 0 \\ -i\alpha & \tilde{\epsilon}_0 & 0 \\ 0 & 0 & \epsilon_{zz} \end{pmatrix}, \quad \boldsymbol{\mu}_F(\beta) = \begin{pmatrix} 1 & i\beta & 0 \\ -i\beta & 1 & 0 \\ 0 & 0 & \mu_{zz} \end{pmatrix}.$$

Mathematical Foundation (3/3)

- Since the tensors change with the material of the layers, we can define their contents in a piecewise fashion:

$$\epsilon(z) = \begin{cases} \epsilon_A(\delta, \varphi_1), & 0 < z < L; \\ \epsilon_F(\alpha), & L < z < 1 - L; \\ \epsilon_A(\delta, \varphi_2), & 1 - L < z < 1. \end{cases} \quad \mu(z) = \begin{cases} \mu_A, & 0 < z < L; \\ \mu_F(\beta), & L < z < 1 - L; \\ \mu_A, & 1 - L < z < 1. \end{cases}$$

Tensor Hermiticity

- If the tensors are equal to their conjugate transpose, they are **Hermitian**.

$$\epsilon(z) = \epsilon^*(z) \quad \text{and} \quad \mu(z) = \mu^*(z)$$

- If both tensors are Hermitian, the system is Hermitian.
- If either tensor is not Hermitian (non-Hermitian), then the system is non-Hermitian.

Wave Propagation

- Consider an electromagnetic wave $\{\mathbf{E}, \mathbf{H}\}$ defined by:

$$\mathbf{E} = [E_x(z), E_y(z), 0]^T e^{-i\omega t} \quad \mathbf{H} = [H_x(z), H_y(z), 0]^T e^{-i\omega t}$$

- Maxwell's equations for the curl of an electric and magnetic field yield:

$$\nabla \times \mathbf{E} = \frac{i\omega}{c} \mu \mathbf{H} \quad \nabla \times \mathbf{H} = -\frac{i\omega}{c} \epsilon \mathbf{E}$$

Forming the ODE System

- Leveraging the physical properties, Maxwell's equations reduce to a simple ODE system:

$$Q \frac{d}{dz} \mathbf{E}(z) = \frac{i\omega}{c} \mu(z) \mathbf{H}(z), \quad Q \frac{d}{dz} \mathbf{H}(z) = -\frac{i\omega}{c} \epsilon(z) \mathbf{E}(z) \quad Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

- We can express this ODE system in vector form with block matrices:

$$\Phi(z) = \begin{pmatrix} \mathbf{E}(z) \\ \mathbf{H}(z) \end{pmatrix}, \quad Q = \begin{pmatrix} Q & 0 \\ 0 & Q \end{pmatrix}, \quad \mathcal{A}(z) = i \begin{pmatrix} 0 & \mu(z) \\ -\epsilon(z) & 0 \end{pmatrix}.$$

$$\mathcal{A}(z+1) = \mathcal{A}(z) \quad \frac{d}{dz} \Phi(z) = \omega Q^{-1} \mathcal{A}(z) \Phi(z)$$

Analytical Solution of the ODE System

- The solution ends up being a 4x4 matrix we call the "transfer matrix". It relates frequencies at different points in the crystal:

$$\Phi(z_2) = \mathcal{T}(\omega; z_2 - z_1) \Phi(z_1)$$

- By *Bloch's Theorem*, we know a wave shifted by the fundamental period obtains a phase factor:

$$\forall k \in [-\pi, \pi] \quad \Phi(k; z + 1) = e^{ik} \Phi(k; z)$$

Preparing the Computational Problem

- We form the unit cell transfer matrix from the constituent transfer matrices:

$$\mathcal{M}(\omega) = \mathcal{T}_A(\omega, \varphi_1; L) \mathcal{T}_F(\omega; 1 - L) \mathcal{T}_A(\omega, \varphi_2; L)$$

- For each k in the Brillouin zone, eigenfrequencies exist where the phase factor given by Bloch's theorem is an eigenvalue of the unit cell transfer matrix:

$$\det(\mathcal{M}(\omega) - e^{ik} I) = 0 \quad \lambda = e^{ik}$$

Defining The Band Structure

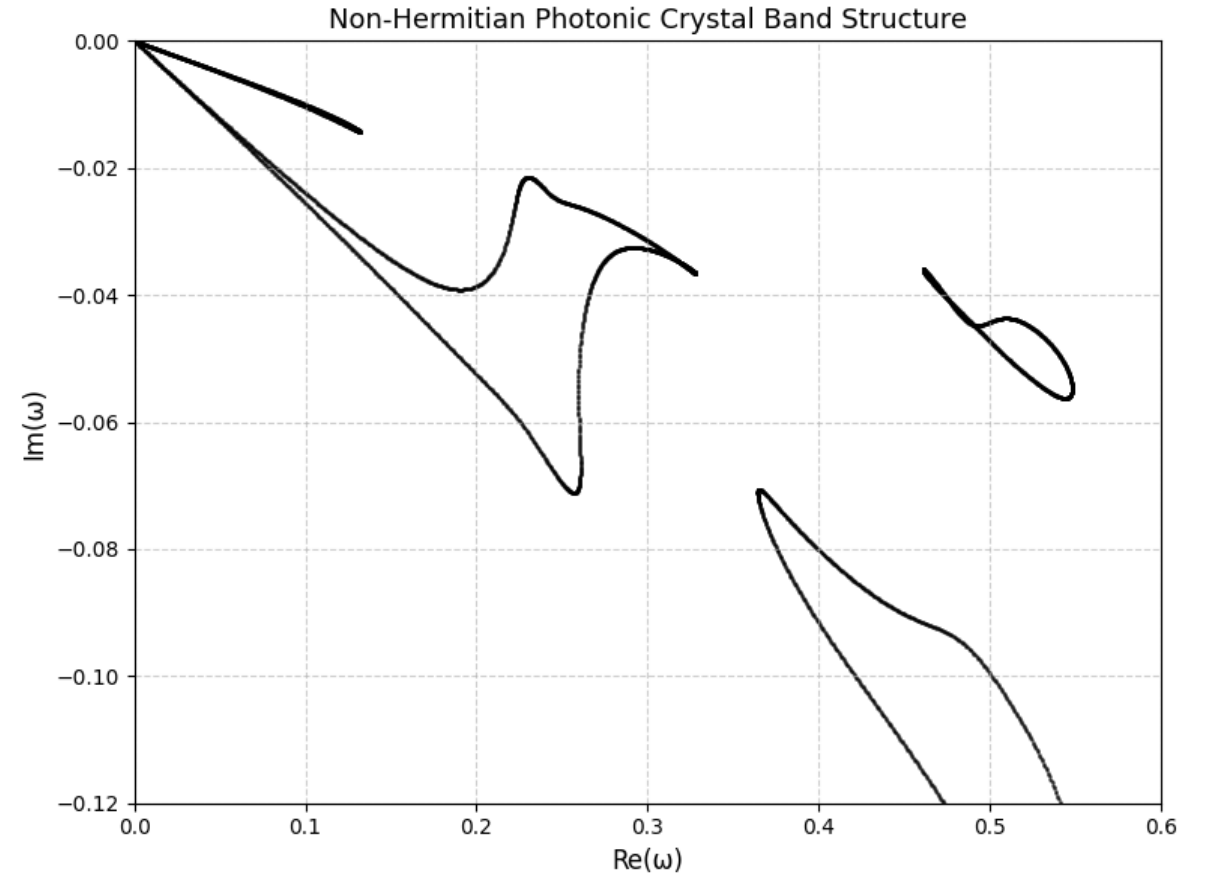
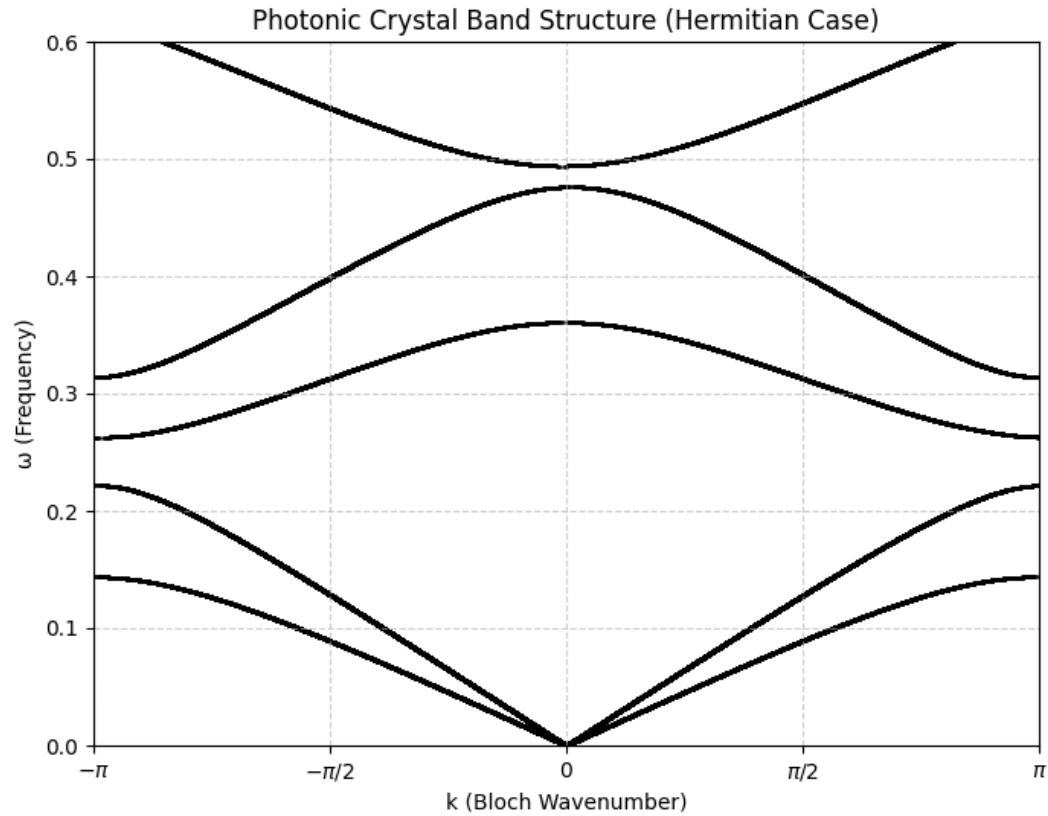
- For the Hermitian case, the band structure is the set of eigenfrequency-phase factor pairs described by:

$$\{\omega_n(k)\}_{n=1}^{\infty} \quad \omega_n(k) : [-\pi, \pi] \rightarrow \mathbb{R}$$

- For the non-Hermitian case, we add an imaginary component to elements of the ferromagnetic permittivity tensor, forcing the system to become non-Hermitian. Then the band structure is defined by loops of complex eigenvalues:

$$\{\omega_n(k)\}_{n=1}^{\infty} \quad \omega_n(k) : [-\pi, \pi] \rightarrow \mathbb{C}$$

Examples of Band Structures



Computing Band Structures Numerically

Algorithm 1 Calculation of Hermitian Band Structure

Require: Set of real frequencies Ω , numerical tolerance ϵ .

Ensure: Set of valid dispersion pairs $B = \{(k, \omega)\}$.

```
1:  $B \leftarrow \emptyset$ 
2: for each  $\omega \in \Omega$  do
3:   Compute unit cell transfer matrix  $\mathcal{M}(\omega)$ 
4:   Calculate the eigenvalues  $\{\lambda_j\}$  of  $\mathcal{M}(\omega)$ 
5:   for each  $\lambda_j$  do
6:     if  $||\lambda_j| - 1| < \epsilon$  then                                ▷ Filter for propagating waves
7:        $k \leftarrow \text{Re}(-i \ln \lambda_j)$                             ▷ Extract Bloch wavenumber
8:        $B \leftarrow B \cup \{(k, \omega)\}$ 
9:     end if
10:  end for
11: end for
12: return  $B$ 
```

Algorithm 2 Calculation of Non-Hermitian Band Structure

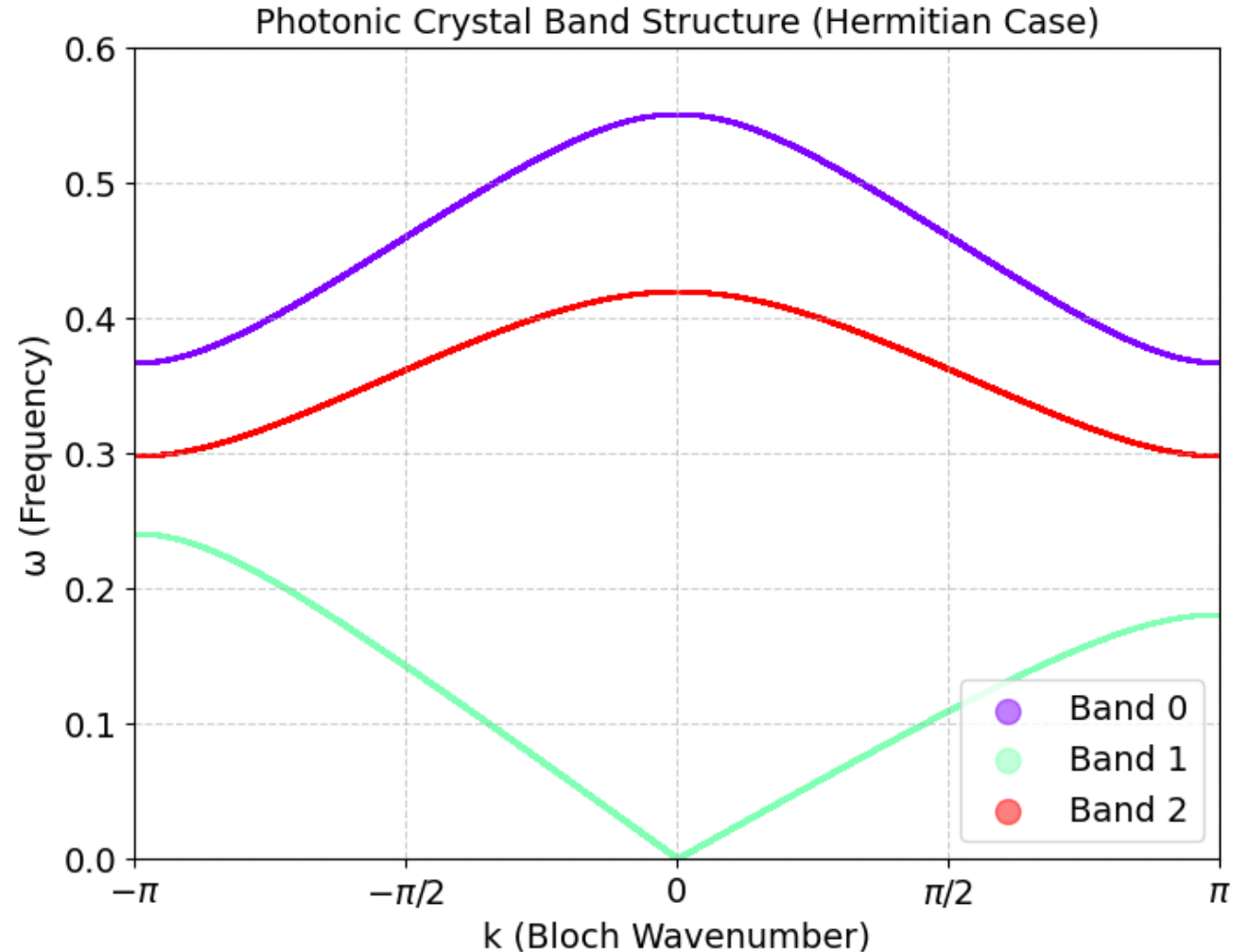
Require: Set of momenta $\mathcal{K}$ in the Brillouin zone $[-\pi, \pi]$.

Ensure: Set of valid dispersion pairs $B = \{(k, \omega)\}$.

```
1:  $B \leftarrow \emptyset$ 
2: for each  $k \in \mathcal{K}$  do
3:   Define characteristic function  $f(\omega) := \det(\mathcal{M}(\omega) - e^{ik}I)$ 
4:   Numerically solve  $f(\omega^*) = 0$  to find complex eigenfrequencies  $\omega^*$ 
5:   for each converged root  $\omega^*$  do
6:      $B \leftarrow B \cup \{(k, \omega^*)\}$ 
7:   end for
8: end for
9: return  $B$ 
```

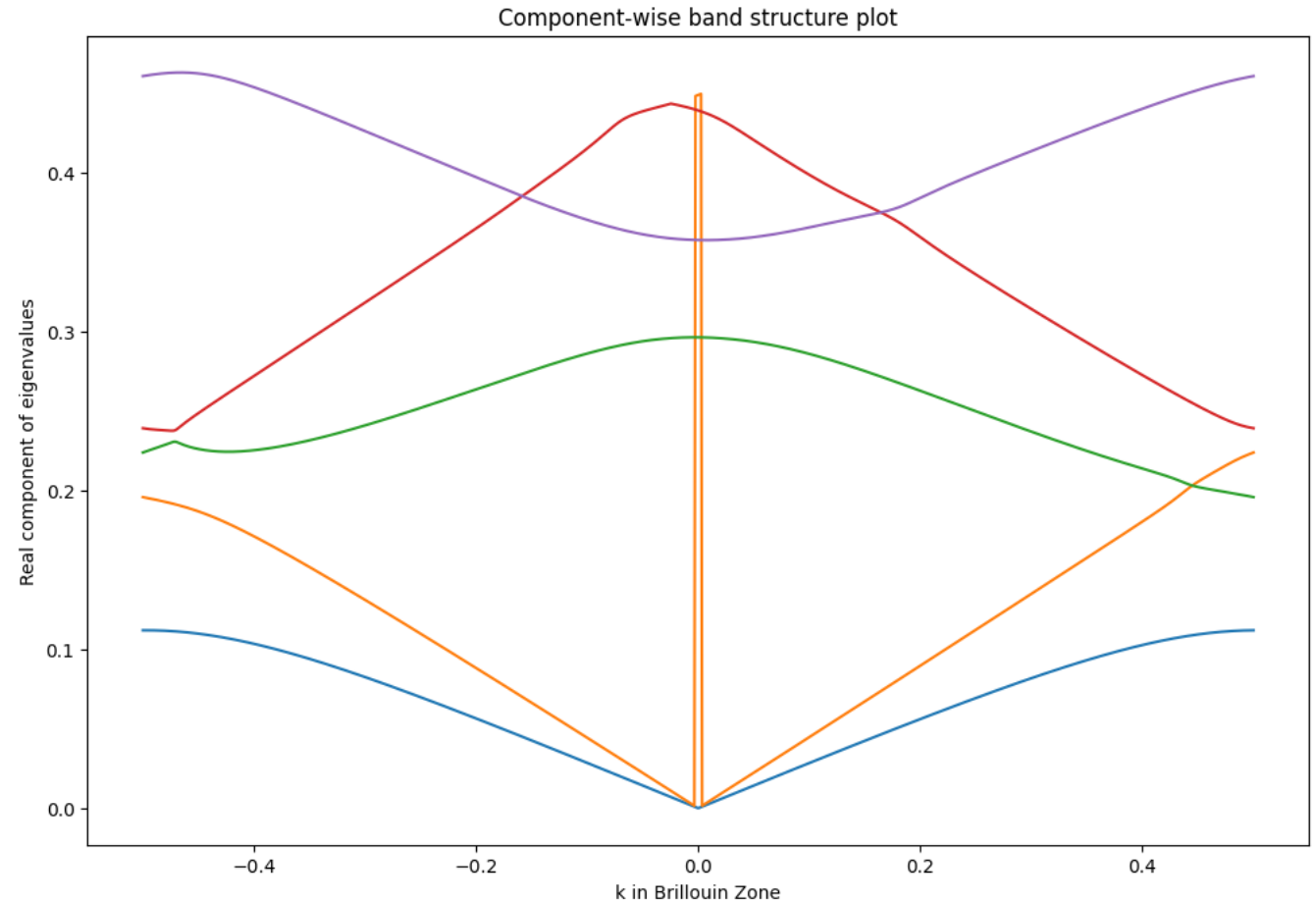
Identifying Band Structures

- Using an algorithm that leverages KD trees, we seek to assign each discrete point to a given band:



Identify Band Structures Ctd.

- For the non-Hermitian cases, we can separate the real and imaginary parts of the eigenvalues and try to connect those separately.
- This method still has limitations, e.g. numerical errors and overlapping bands.



Future Work

- Perfect band detection algorithms for both Hermitian and non-Hermitian cases.
- Analyze dataset:
 - Calculate winding numbers.
 - Relate band properties to physical crystal properties.
 - Use machine learning to classify bands with CNNs or MLPs.

Thank you!